Suppressing Sycophancy Cascades and Trauma Lock-In in Autonomous Financial Multi-Agent LLMs: Bounded Adversarial Deliberation, Exponential Time-Decay Episodic Memory, and Foundation Model OOD Veto

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ORCID: 0009-0009-8292-5085 | PyPI Package: `pip install lumina-quant` (v6.6.5) | Repository: `github.com/ramuza0113-sketch/lumina-ai-hedge-fund`

ABSTRACT

Autonomous financial decision-making architectures powered by Large Language Model (LLM) ensembles and episodic verbal reinforcement learning (Reflexion / Retrieval-Augmented Generation) exhibit three structural cognitive pathologies when deployed across non-stationary market regimes: (1) **Sycophancy Cascades (Herding Amplification)**, wherein sequential multi-agent debate induces conformist token generation that collapses cross-sectional stance variance and amplifies false-positive entries during speculative market tops; (2) **Episodic Trauma Lock-In (Learned Helplessness in Lexical Memory)**, where retrieving historical stop-loss reflections via time-invariant BM25 similarity allows severe drawdown clusters from past bear markets to permanently dominate top-k context windows even after macroeconomic recovery, freezing capital in perpetual `HOLD` states; and (3) **Time-Series Foundation Model Out-of-Distribution (OOD) Hallucination**, where pre-trained zero-shot forecasters (Google TimesFM, Amazon Chronos) extrapolate bullish price trajectories on structurally insolvent equities experiencing institutional order-book distribution.

Building upon the deterministic microstructure and factor ontology invariants established in Hashimoto (2026a), we formulate a unified mathematical framework to eliminate all three cognitive failure modes. First, we construct a **Parallel Blind Orthogonal Evaluation Protocol** paired with a **Bounded Adversarial Debate Perturbation Operator** ($|\Delta S_i| \leq \kappa_{\max}$) and an **HTTP 429 Degraded Defense-Only Circuit Breaker** that guarantees zero blind execution during LLM inference blackouts. Second, we introduce a **Time-Decayed Compound-Token BM25 Episodic Memory Kernel** featuring an exponential half-life attenuation factor $w(\Delta t) = 2^{-\Delta t / \tau_{1/2}}$ ($\tau_{1/2} = 30.0$ days) and derive the closed-form **Trauma Horizon Escape Inequality** governing optimal memory forgetting across regime transitions. Third, we establish a **Deterministic Chief Risk Officer (CRO) Absolute Veto Operator** that overrides foundation-model bullish hallucinations whenever fundamental solvency and 10-level Order Flow Imbalance (OFI) breach structural safety bounds. Empirical ablation across a 5-year multi-asset dataset (2021–2026) and 20 automated chaos-engineering unit tests confirms a 0.0% trauma lock-in rate, a 71.4% reduction in herd-driven false breakouts, and 100% immunity to foundation-model OOD hallucinations.

Keywords: Multi-Agent LLM Systems, Anti-Sycophancy Deliberation, Episodic Memory, Exponential Time Decay, BM25 Retrieval, Learned Helplessness, Time-Series Foundation Models, Out-of-Distribution Veto, Quantitative Finance
JEL Classification: G11, G12, G14, G17, C53, C58, C63

1. INTRODUCTION AND MOTIVATION

The integration of Large Language Models (LLMs) into quantitative finance has shifted from static sentiment classification toward stateful, autonomous multi-agent investment committees capable of synthesizing corporate filings, macroeconomic indicators, and technical factor matrices (Yang et al., 2020; Liu et al., 2020). In a companion study (Hashimoto, 2026a), we resolved three foundational input- and execution-layer distortions: semantic polarity inversion in raw numeric factors via Schmitt-Trigger hysteresis ontologies, limit order book horizon truncation bias via scroll-out resistant Deep Order Flow Imbalance (OFI), and jump-slippage ruin in asymmetric pyramiding. However, even when every quantitative input fed into

an LLM ensemble is canonicalized and mathematically bounded, the *internal deliberation and episodic memory mechanics* of generative agents introduce a second class of failure modes during live multi-year operation.

First, generative language models trained with Reinforcement Learning from Human Feedback (RLHF) exhibit strong *sycophancy bias*—a statistical propensity to align generated responses with the perceived stance of preceding context tokens (Perez et al., 2022; Sharma et al., 2023). In sequential multi-agent trading architectures (where Agent B observes Agent A's transcript before speaking), an initial high-confidence bullish narrative by a momentum persona triggers a sycophancy cascade across value and macro personas, collapsing epistemic diversity precisely when contrarian skepticism is most needed.

Second, to enable continuous self-improvement without expensive weight fine-tuning, modern agentic systems employ verbal reinforcement learning (*Reflexion*; Shinn et al., 2023), storing post-mortem reflections of losing trades in a full-text or vector database and retrieving top-k lessons via lexical similarity (Robertson & Zaragoza, 2009) prior to new trade evaluations. We prove both theoretically and empirically that in cyclical financial markets, unweighted lexical retrieval induces **Episodic Trauma Lock-In**: following a concentrated volatility shock (such as the 2022 technology drawdown or the August 2024 Nikkei flash crash), high-density stop-loss reflections saturate the lexical index. Because standard BM25 scoring is time-invariant, these 90-to-180-day-old trauma records continue to outrank fresh post-recovery reflections, causing the investment council to reject valid breakout setups indefinitely—an artificial analog of biological learned helplessness.

Third, hybrid architectures increasingly incorporate pre-trained Time-Series Foundation Models (TSFMs) such as Google TimesFM (Das et al., 2024) and Amazon Chronos (Ansari et al., 2024) for zero-shot price trajectory forecasting. Because TSFMs condition solely on univariate price patches without access to balance-sheet solvency or limit order book depth, a short-squeeze bounce in a structurally bankrupt equity frequently elicits a high-probability `DOUBLE_BULLISH` forecast. Without a deterministic cross-modal veto, LLM agents anchor on the foundation model's numerical authority and allocate capital into imminent dilution or delisting traps.

2. BOUNDED ANTI-SYCOPHANCY DELIBERATION AND DEGRADED CIRCUIT BREAKERS

2.1 Parallel Blind Orthogonal Scoring vs. Sequential Herding

Consider an investment committee comprising $N = 7$ heterogeneous analyst personas $\mathcal{A} = \{a_1, \dots, a_7\}$ representing distinct economic utility functions (Intrinsic Value, Deep Margin-of-Safety, Disruptive Innovation, Secular Contrarian Short, Reflexive Macro, CANSLIM Momentum, and Microstructure Tape Reading), overseen by a Chief Risk Officer (a_{CRO}) and a Chief Investment Officer (a_{CIO}). Let $\mathcal{F}_t^{\text{can}}$ denote the canonical Schmitt-Trigger factor ontology state at time t as defined in Hashimoto (2026a).

In a naive sequential deliberation graph, agent a_i conditions on the prior outputs $\{y_1, \dots, y_{i-1}\}$, yielding conditional score variance $\mathbb{V}[S_i \mid y_{1:i-1}] = (1 - \rho_{\text{syc}}^2)^i \sigma_0^2$, which decays exponentially toward zero as the sycophancy correlation $\rho_{\text{syc}} \in (0, 1)$ compounds across turns. To eliminate sequential anchoring, Lumina Capital enforces **Stage 1 Parallel Blind Evaluation**: each agent $a_i \in \mathcal{A}$ independently maps its assigned domain subset $\mathcal{F}_{t,i}^{\text{can}} \subset \mathcal{F}_t^{\text{can}}$ and time-decayed episodic memory $\mathcal{M}_{t,i}$ into an initial score $S_i^{(0)} \in [0, 100]$ and directional stance $d_i^{(0)} \in \{-1, 0, +1\}$ with zero cross-agent visibility:

$$I(S_i^{(0)}; S_j^{(0)} \mid \mathcal{F}_t^{\text{can}}, \mathcal{M}_t) = 0 \quad \forall i \neq j \in \{1, \dots, N\}$$

2.2 Adversarial Extremum Pairing and Bounded Reversal Operator

Following parallel blind scoring, rather than exposing all agents to an unstructured group chat, the orchestrator isolates the two maximum-divergence personas—the strongest proponent $a^+ = \arg\max_i S_i^{(0)}$ and the strongest dissenter $a^- = \arg\min_j S_j^{(0)}$—for a single structured adversarial rebuttal $\mathcal{D}(a^+, a^-)$. To prevent rhetorical persuasion from

overriding quantitative fundamentals during synthesis, we constrain post-debate score adjustments $\Delta S_i = S_i^{(1)} - S_i^{(0)}$ via a piece-wise **Bounded Debate Perturbation Operator**:

> **Theorem 1 (Anti-Sycophancy Consensus Stability Bound).** *Let $\bar{S}_W^{(0)} = \sum_{i=1}^{N} w_i S_i^{(0)}$ ($\sum w_i = 1, w_i > 0$) denote the regime-weighted blind consensus score, and let $N_{\text{bull}} = \sum_{i=1}^{N} \mathbb{I}(S_i^{(0)} \geq \theta_{\text{bull}})$ denote the initial bullish quorum count. Under the Lumina bounded rebuttal rule, individual agent score revisions ΔS_i are restricted to:*
>
> $$|\Delta S_i| \leq \kappa(N_{\text{bull}}) \equiv \begin{cases} \kappa_{\text{minor}} = 3.0 \text{ pts}, & \text{if } N_{\text{bull}} < \lceil N/2 \rceil \text{ (Minority Bull Dissent)} \\ \kappa_{\text{major}} = 8.0 \text{ pts}, & \text{if } N_{\text{bull}} \geq \lceil N/2 \rceil \text{ (Quorum Established)} \end{cases}$$
>
> *Consequently, whenever an asset fails to achieve an initial fundamental quorum ($N_{\text{bull}} \leq 3$ out of 7), no rhetorical hallucination during adversarial debate can elevate the post-debate consensus $\bar{S}_W^{(1)}$ by more than $\kappa_{\text{minor}} = 3.0$ points, strictly preventing sycophancy-induced threshold crossing whenever $\bar{S}_W^{(0)} < \Theta_{\text{exec}} - \kappa_{\text{minor}}$.*

2.3 HTTP 429 Fast-Fail Circuit Breaker & Degraded Defense-Only Mode

In live production, external LLM inference endpoints periodically suffer rate-limit exhaustion (HTTP 429 `RESOURCE_EXHAUSTED`) or transient network blackouts. If an autonomous trading engine substitutes simple deterministic rule-based scores during an LLM outage and continues executing `BUY` orders, it risks entering unvetted positions without qualitative risk auditing. We formalize a two-state **Inference Circuit Breaker** $\mathcal{C}_t \in \{\text{CLOSED}, \text{OPEN}\}$: upon any HTTP 429 or timeout exception, $\mathcal{C}_t$ transitions immediately to OPEN for a cooldown horizon $\Delta t_{\text{cool}} = 60$ s (halting cascading retry storms) and sets the portfolio synthesis flag `degraded_mode = True`. In Degraded Defense-Only Mode, stop-loss monitoring, stock-split scaling, and trailing-stop liquidations remain 100% operational, but new capital allocation is mathematically clamped to zero:

$$\mathcal{C}_t = \text{OPEN} \implies \text{Action}_t \equiv \text{HOLD}, \quad w_{\text{alloc}}^{(t)} \equiv 0.0\% \quad \forall \bar{S}_W^{(0)} \in [0, 100]$$

3. EXPONENTIAL TIME-DECAY EPISODIC MEMORY AND OPTIMAL TRAUMA FORGETTING

3.1 Compound-Token FTS5 Lexical Indexing and Single-Writer Concurrency

Every closed position generates a structured post-mortem record $m_j = (s_j, a_i, r_j, \text{rule}_j, \tau_j, t_j)$ containing ticker symbol s_j (e.g., `3556.T`, `NVDA`), agent identifier a_i, realized return r_j, exit trigger rule_j (e.g., `trailing_stop_sweep`, `gamma_call_wall`), natural-language reflection τ_j, and creation timestamp t_j. Standard SQLite FTS5 tokenizers strip punctuation characters (`-`, `_`, `.`), fragmenting domain-critical financial identifiers such as `3556.T` or `trailing_stop_sweep` into ambiguous sub-tokens. We configure a custom FTS5 lexical index with explicit symbol preservation (`tokenize="unicode61 tokenchars '-._'"`) and serialize all concurrent multi-threaded agent reflection writes through an asynchronous **Single-Writer Batch Queue**, achieving zero SQLite `database is locked` contention under 64-thread burst liquidations.

3.2 Time-Decayed BM25 Retrieval Kernel and Trauma Escape Theorem

Let $\text{BM25}(q, m_j) > 0$ denote the raw Okapi BM25 lexical relevance score between current market query q and historical reflection m_j recorded at timestamp $t_j \leq t$:

$$\mathrm{BM25}(q, m_j) = \sum_{v \in q} \mathrm{IDF}(v) \cdot \frac{f(v, m_j) \cdot (k_1 + 1)}{f(v, m_j) + k_1 \left(1 - b + b \frac{|m_j|}{\bar{L}}\right)}$$

To prevent ancient drawdown clusters from permanently dominating top-k memory retrieval after a market regime shift, we weight each candidate memory by an exponential radioactive decay kernel with half-life $\tau_{1/2} > 0$ (calibrated to $\tau_{1/2} = 30.0\ \mathrm{days}$, matching monthly macroeconomic release cycles):

$$R(q, m_j, t) = \mathrm{BM25}(q, m_j) \cdot w(t - t_j), \qquad w(\Delta t) = 2^{-\Delta t / \tau_{1/2}} = \exp\left(-\frac{\ln 2}{\tau_{1/2}} \Delta t\right)$$

> **Theorem 2 (Episodic Trauma Horizon Escape Inequality).** *Consider a severe historical stop-loss trauma memory m_{trauma} recorded at t_{trauma} with lexical relevance $B_{\text{trauma}} = \mathrm{BM25}(q, m_{\text{trauma}})$, and a fresh post-recovery reflection m_{fresh} recorded at $t_{\text{fresh}} \approx t$ with relevance $B_{\text{fresh}} = \mathrm{BM25}(q, m_{\text{fresh}})$. Under time-invariant retrieval ($\tau_{1/2} \to \infty$), if $B_{\text{trauma}} \geq B_{\text{fresh}}$, the trauma memory outranks the fresh lesson indefinitely ($\forall t > t_{\text{trauma}}$). Under the exponential decay kernel $w(\Delta t) = 2^{-\Delta t / \tau_{1/2}}$, the fresh reflection strictly outranks the historical trauma ($R(q, m_{\text{fresh}}, t) > R(q, m_{\text{trauma}}, t)$) for all elapsed horizons $\Delta t = t - t_{\text{trauma}}$ satisfying:*
>
> $$\Delta t > \Delta t^{\ast} \equiv \tau_{1/2} \log_2 \left(\frac{B_{\text{trauma}}}{B_{\text{fresh}}} \right)$$
>
> *In particular, for identical keyword matches ($B_{\text{trauma}} = B_{\text{fresh}}$), a 90-day-old crash trauma ($\Delta t = 90\ \mathrm{d} = 3\tau_{1/2}$) is attenuated to $w(90) = 2^{-3} = 0.125$ (87.5% attenuation), guaranteeing that current-regime reflections ($w(0) = 1.000$) dominate top-k agent context with an $8:1$ priority ratio.*

4. DETERMINISTIC CRO VETO AGAINST FOUNDATION MODEL OOD HALLUCINATIONS

To integrate zero-shot forecasts from Google TimesFM ($\hat{r}_{\text{TFM}}$) and Amazon Chronos ($\hat{r}_{\text{Chr}}$) without exposing the portfolio to Out-of-Distribution (OOD) price extrapolations on structurally distressed equities, we define a cross-modal structural distress predicate $\mathcal{D}_{\text{struct}}(s, t)$ combining Piotroski fundamental health ($F_s \in \{0, \dots, 9\}$), Return on Equity (ROE_s), long-term trend alignment ($P_{s,t}$ vs. $\mathrm{SMA}_{200,s}$), and 10-level scroll-out resistant Order Flow Imbalance ($\mathrm{OFI}_{10}^{(s,t)}$ from Hashimoto, 2026a):

$$\mathcal{D}_{\text{struct}}(s, t) \equiv \mathbb{I}(F_s < 6 \ \wedge \ \mathrm{ROE}_s < 0\% \ \wedge \ P_{s,t} < \mathrm{SMA}_{200,s}) \ \vee \ \mathbb{I}\left(\mathrm{OFI}_{10}^{(s,t)} \leq -40.0\right)$$

Whenever the foundation model ensemble emits a high-conviction bullish signal ($\mathrm{Status}_{\text{TSFM}} = \mathrm{DOUBLE_BULLISH}$, $\hat{r}_{\text{ens}} > 0$) on an asset satisfying $\mathcal{D}_{\text{struct}}(s, t) = 1$, the Chief Risk Officer (a_{CRO}) triggers an unconditional **Deterministic OOD Veto** ($\mathrm{Veto}_{\text{CRO}} = \mathrm{True}$), overriding any bullish analyst scores and forcing $w_{\text{alloc}}^{(s,t)} = 0.0\%$.

5. FORMAL VERIFICATION AND 5-YEAR EMPIRICAL ABLATION (2021–2026)

5.1 Automated Chaos Engineering & Cognitive Invariant Verification

All mathematical invariants derived in Sections 2–4 are implemented in the open-source Python package `lumina-quant` (v6.6.5) and verified via an offline `pytest` suite executing without external API keys:

Verification Test ID	Adversarial Scenario & Mathematical Invariant Tested	Empirical Measurement	Status
`test_3_sqlite_fts5`	8-thread $\times$ 8 burst concurrent reflection writes (64 records) + compound token recall (`3556.T` , `trailing_stop_sweep`)	0 SQLite lock errors; 100% compound token recall	**PASS**
`test_8_reflexion_decay`	Identical BM25 keyword collision between fresh lesson ($\Delta t = 0\text{d}$) and bear-market trauma ($\Delta t = 90\text{d} = 3\tau_{1/2}$)	Fresh $w = 1.000$ vs. 90d Old $w = 0.125$ (Exact $8:1$ ratio)	**PASS**
`test_9_circuit_breaker`	Simulated HTTP 429 quota exhaustion with artificially high fallback analyst scores (88 pts across all 7 personas)	`degraded_mode=True` ; 0 blind `BUY` orders (`alloc=0%`)	**PASS**
`test_12_tsfm_ood_veto`	Adversarial `DOUBLE_BULLISH` (+12.5%) TSFM forecast injected on distressed equity ($F = 2, \text{ROE} = -35\%, \text{OFI} = -55$)	CRO Absolute Veto triggered; `action='HOLD'` , `alloc=0%`	**PASS**
`test_18_sigkill_hydration`	Abrupt process termination (`kill -9` / `SIGKILL`) wiping in-memory cooldowns followed by SQLite state hydration	100% restoration of 14-day anti-churn cooldown state	**PASS**

5.2 5-Year Multi-Asset Empirical Ablation Study (2021–2026)

Table 2 reports the 5-year out-of-sample ablation performance across US equities, Tokyo Stock Exchange equities, and spot precious metals (2021–2026), isolating the incremental contribution of Parallel Anti-Sycophancy Deliberation, $\tau_{1/2} = 30\text{d}$ Time-Decayed Episodic Memory, and the CRO OOD Veto:

Architectural Configuration	Herd False-Breakout Rate	Post-Crash Trauma Freeze Days	TSFM OOD Loss Incidence	Out-of-Sample Profit Factor / DSR
1. Baseline Sequential LLM Debate + Raw BM25 *(Unbounded sequential chat, $\tau_{1/2} = \infty$, no TSFM veto)*	29.4%	84.0 days *(Chronic lock-in)*	9.8% of trades	PF 1.41 DSR 61.8%
2. + Parallel Blind Scoring & Bounded Rebuttal *(Theorem 1 enabled; $\kappa_{\text{minor}} = 3.0$ pts, $\tau_{1/2} = \infty$)*	11.2% *(-61.9% reduction)*	79.5 days *(Still locked by BM25)*	7.4% of trades	PF 1.72 DSR 84.3%
3. + Exponential Time-Decay Memory ($\tau_{1/2} = 30\text{d}$) *(Theorem 1 + Theorem 2 enabled; no TSFM OOD veto)*	9.6%	0.0 days *(Optimal re-entry)*	6.1% of trades	PF 1.96 DSR 94.2%
4. Full Proposed Architecture (`Lumina v6.6.5`) *(Theorems 1 & 2 + CRO OOD Veto + 429 Circuit Breaker)*	8.4% *(-71.4% reduction)*	0.0 days *(Zero trauma lock-in)*	0.0% *(100% vetoed)*	PF 2.10 / 2.48 DSR 99.14%

6. CONCLUSION AND AI USAGE DISCLOSURE

We have demonstrated that transforming multi-agent LLMs from fragile conversational demonstrations into institutional-grade quantitative decision engines requires bounding not only their market microstructure inputs (Hashimoto, 2026a) but also their internal cognitive dynamics. By enforcing parallel blind orthogonal scoring with bounded debate perturbations (Theorem 1), attenuating episodic BM25 reflections via a 30-day exponential half-life kernel (Theorem 2), and interposing a deterministic CRO veto against foundation-model OOD hallucinations, `lumina-quant` eliminates sycophancy cascades and post-drawdown trauma paralysis while maintaining full offline reproducibility.

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